SCHEDULE 'A'

TIRUMALA TIRUPATI DEVASTHANAMS

BILL OF QUANTITIES

NIT No.92 /SE-I/TTD/2026-27 **Period : 12 months**

NAME OF WORK: "Providing fencing all around 12 Ac and 20 Ac land and construction of new sheads for storage of Paddy straw and Silage shed at S.V. Gosamrakshana Shala, TTD, Tirupati".

S.No.	Estimate Quantity (only Figures)	Item Detailed Specification Description	Work Type eg. Earth Work, Electrical works.. etc (upto 200 Characters)	Item Short Description (upto 100 Characters)	APSS / Morth Cl. Number (upto 200 Characters)	Rate (INR) (Upto 2 Decimals)	UOM (upto 50 Characters)	Amount (INR) (Upto 2 Decimals)
1	1950.00	Taking out existing CC interlocking paver blocks from footpath / central verge, including removal of rubbish etc., disposal of unserviceable material to the dumping ground, for which payment shall be made separately and stacking of serviceable material within 50 metre lead as per direction of Engineer-in-Charge	Taking out existing CC interlocking paver blocks from footpath / central ve	Taking out existing CC interlocking paver blocks f	NA	122.22	1Sqm	238329.00
2	600.00	Earthwork excavation for foundations and depositing on bank for all lifts and with an initial lead of 10 m including all operational, incidental, labour charges such as shoring, sheeting, planking, strutting etc., complete for finished item of work. Ordinary soil - Mechanical means up to 3 m depth. Note: Cost of dewatering up to 5% may be added, where required, up to 10% of labour cost. Assessment for dewatering shall be made as per site conditions.	Earthwork excavation for foundations and depositing on bank for all lifts	Earthwork excavation for foundations and deposit	308	70.25	1Cum	42150.00

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3	600.00	Filling with useful available excavated earth (excluding rock) in trenches, sides of foundations and basement with initial lead in layers not exceeding 15 cm thick, consolidating each deposited layer by watering and ramming including cost and conveyance of water to work site and all operational, incidental, labour charges, hire charges of T&P etc., complete for finished item of work. Note: Cost of transportation of good quality earth has not been included. Only labour for carrying carted earth with a lead of 50 m to the foundation pits has been taken in the rate. The cost of carted earth / sand may be worked out separately if the same is not available from adjoining area.	Filling with useful available excavated earth (excluding rock) in trenches,	Filling with useful available excavated earth (exc	308	52.10	1Cum	31260.00
4	142.00	Filling of foundations and basement with best river sand in layers of not more than 15cm thick, each well watered to saturation and consolidated including cost and conveyance of all materials, incidental charges etc., complete.	Filling of foundations and basement with best river sand in layers of not m	Filling of foundations and basement with best rive	309 310	982.65	1Cum	139536.30
5	295.00	Filling of foundations and basement with good gravel in layers of not more than 15 cms. thick each well watered to saturation and consolidated including cost and conveyance of all materials , incidental charges etc., complete.	Filling of foundations and basement with good gravel in layers of not more	Filling of foundations and basement with good grav	309 310	579.64	1Cum	170993.80

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6	137.00	Cement Concrete (1:4:8) using 40 mm (0.54 cum), 20mm (0.27 cum) and 12 mm (0.09 cum) nominal size hard broken granite crushed metal from any source conforming to the relevant ISI standards for foundations etc., laid in layers of not more than 15 cms thick well compacted including curing, machine mixing, cost and conveyance of all materials etc., complete.	Cement Concrete (1:4:8) using 40 mm (0.54 cum), 20mm (0.27 cum) and 12 mm (	Cement Concrete (1:4:8) using 40 mm (0.54 cum), 20	NA	5374.66	1Cum	736328.42
7	8.00	Cement Concrete (1:2.5:5) Grade M15 using 20 mm (0.54 cum) and 12 mm (0.36 cum) nominal size hard broken granite crushed metal from any source conforming to the relevant ISI standards for foundations etc., laid in layers of not more than 15 cms thick well compacted including curing, machine mixing, cost and conveyance of all materials etc., complete. Note: Measurement boxes shall be used for measuring the quantities of cement, aggregate and water required for the given concrete mix.	Cement Concrete (1:2.5:5) Grade M15 using 20 mm (0.54 cum) and 12 mm (0.36	Cement Concrete (1:2.5:5) Grade M15 using 20 mm (0	NA	6224.40	1Cum	49795.20

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8		Supply and placing of M 25 Design Mix Concrete corresponding to IS 456 using concrete mixer using of 20 mm HBG crushed metal 0.530 cum (707 kg), 10 mm HBG crushed metal, 0.362 cum (471 kg), sand 0.407 cum (616 kg.) and cement 380 kg with water cement ratio of 0.48 (182.40 ltrs/ cum of CC) etc., to site, including sales and other taxes on all materials, operational, incidental and labour charges such as machine mixing, vibrating, laying concrete, curing etc., complete with cement from standard suppliers approved by the department, including centering and scaffolding charges, using steel scaffolding pipes, jack props, wallers, foot plates, brackets, steel centering plates, etc., and excluding cost of steel and its fabrication charges for finished item of work (APSS No. 402). Note: Measurement boxes shall be used for measuring the quantities of cement, aggregate and water required for the given concrete mix if mixing is done by volume mixing.	Supply and placing of M 25 Design Mix Concrete corresponding to IS 456 us	Supply and placing of M 25 Design Mix Concrete c	402	0.00	NA	0.00

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0		Note : The design mix shall be got designed by furnishing the information required as per clause 19.1.1 & 19.1.2 of IS 456-2000 by the contractor at his own cost and get approved from the competent authority before execution. The cement considered for above M25 grade concrete is 380 Kg/Cum and the rate is arrived accordingly in the estimate. After the mix is designed and approved, if there is any variation in the quantity of cement, over the cement content adopted in the estimate, the same will be considered and paid/recovered at estimate rate + or - T.P (Either increase or decrease in quantity adopted in the estimate). Payment will not be made for variation in the quantity of other materials such as coarse & Fine aggregates.	Note : The design mix shall be got designed by furnishing the information r	Note : The design mix shall be got designed by fur	402	0.00	NA	0.00
(a)	36.00	- do - For Footings	- do - For Footings	- do - For Footings	402	10808.37	1Cum	389101.32
(b)	8.50	- do - For Columns up to basement/Pedestals	- do - For Columns up to basement/Pedestals	- do - For Columns up to basement/Pedestals	402	11479.77	1Cum	97578.05
(c)	10.50	- do - For Plinth beams	- do - For Plinth beams	- do - For Plinth beams	402	13722.70	1Cum	144088.35

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9	6.00	Providing High Yield Strength deformed (HYSD) / Thermo mechanically treated (TMT) steel bars below 36mm of Fe 500/ Fe-500D / Fe 550 grade (as per IS 1786 - 2008) of different diameters for RCC works, including labour charges for straightening, cutting, bending to required sizes and shapes, placing in position with cover blocks of approved materials and size and tying and lap splicing with binding wire of 18 SWG, forming grills for reinforcement work as per approved designs and drawings, including cost and conveyance of steel bars including all wastages such as over laps, couplings, chairs, spacer bars including cost and conveyance of binding wire, cover blocks and all incidental, operational, labour charges such as cutting, bending, placing in position, tying including sales and other taxes on all materials etc., complete for finished item of work in all floors. (APSS NO.126) (In the absence of 18G binding wire, 2 Nos. of 20 G binding wire in tandem are to be strictly adhered). Note: Unless otherwise fabrication and placement of steel, no pre-payment shall be made against the supply of steel reinforcement.	Providing High Yield Strength deformed (HYSD) / Thermo mechanically treated	Providing High Yield Strength deformed (HYSD) / Th	126/IS 1786	87370.41	1MT	524222.46
10	19.70	Construction of brick masonry in CM (1:6) using well burnt bricks 2nd class or ground moulded (Non- modular or traditional size) (230 X 110 X 70 mm) including curing, cost and conveyance of all materials as directed during execution and complying with standard specifications etc., complete.	Construction of brick masonry in CM (1:6) using well burnt bricks 2nd clas	Construction of brick masonry in CM (1:6) using w	501 504	8377.06	1Cum	165028.08

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11		Plastering in cement mortar using best river sand from approved quarry including cost and conveyance of all materials, screening, curing, rounding of corners, incidental charges as directed during execution complying with standard specification etc., complete for the following proportions.	Plastering in cement mortar using best river sand from approved quarry incl	Plastering in cement mortar using best river sand	901 903	0.00	NA	0.00
(a)	90.00	- do - in CM (1:5) 12 mm thick	- do - in CM (1:5) 12 mm thick	- do - in CM (1:5) 12 mm thick	901 903	320.10	1Sqm	28809.00
(b)	110.00	- do - in CM (1:5) 20 mm thick	- do - in CM (1:5) 20 mm thick	- do - in CM (1:5) 20 mm thick	901 903	481.74	1Sqm	52991.40
12	520.00	Cement Concrete (1:3:6) using 20 mm (0.54 cu.m) and 12 mm (0.36 cu.m) nominal size hard broken granite crushed metal from any source conforming to the relevant ISI standards for foundations etc., laid in layers of not more than 15 cms thick well compacted including curing, machine mixing, cost and conveyance of all materials etc., complete. Note: Measurement boxes shall be used for measuring the quantities of cement aggregate and water required for the given concrete mix.	Cement Concrete (1:3:6) using 20 mm (0.54 cu.m) and 12 mm (0.36 cu.m) nomin	Cement Concrete (1:3:6) using 20 mm (0.54 cu.m) an	NA	5961.28	1Cum	3099865.60
13	1030.00	Hire charges for centring and scaffolding of VCC Vertical walls of plane surface (rate is for both sides centering) including open drains. Note: Centring charges are for both faces of the respective members. The measurement of centring charges shall be in "Sqm" for one face only and the centring charges given are for two faces of centring.	Hire charges for centring and scaffolding of VCC Vertical walls of plane su	Hire charges for centring and scaffolding of VCC V	NA	624.92	1Sqm	643667.60

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14	45000.00	Supply, fabrication and erection of steel structure in position by using MS Tubes & Pipes of all diameters as per IS -1161, including labour charges for fabricating all heavy steel works like trusses, stanchions, heavy beams and girders including cost of welding rods, power changes etc., and cost and conveyance of all materials and including labour charges for fixing heavy iron grills, trusses, stanchions, heavy beams and girders erecting in position and fixing by using chain pulley blocks, derrek pole arrangements and cranes etc., complete and painting two coats with ready mixed synthetic enamel paint in all shades grade-I having VOC (Volatile Organic Compound) content less than 50 grams/ liter approved colour and make, including brushes to have an even shade, over priming coat with ready mixed red oxide metal primer paint Gr-I (total three coats), including cost and conveyance of all materials, as directed during execution complying with standard specifications etc., complete as directed during execution. (TATA / SAIL / VSP / JINDAL) shall be used. Note:- If any size of MS Tubes & Pipes of TATA / SAIL / VSP / JINDAL is not available, then only the filed Engineers' shall permit for usage of APOLLO Brand (or) as specified during execution.	Supply, fabrication and erection of steel structure in position by using M	Supply, fabrication and erection of steel structur	NA	204.65	1Kg	9209250.00

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15	11100.00	Supplying and fixing in position MS Tubes & pipes of all diametres as per IS-1161 of approved design, including labour charges for fabricating all heavy steel works like trusses, stanchions, heavy beams and girders including cost of welding rods, power changes etc., and cost and conveyance of all materials and fixing, necessary cuttings, bending, welding, grinding and finishing smooth, including cost of welding rods, power charges, including painting two coats with synthetic enamel paint grade-I, over a primary coat (Red oxide metal primer) (total three coats) of approved make and colour including cost of making the holes in to RCC or masonry works and making good the same wherever necessary, including labour charges for fixing in position, cost and conveyance of all materials, incidental charges etc., complete.	Supplying and fixing in position MS Tubes & pipes of all diametres as per I	Supplying and fixing in position MS Tubes & pipes	NA	172.05	1Kg	1909755.00
16	24750.00	Supplying and fixing in position MS Ornamental grills for windows and ventilators or for verandah openings using MS Flats, square rods, angles sheets etc., of approved design including necessary cuttings, bending fabrication, welding, grinding and finishing smooth including painting the grills with two coats with synthetic enamel paint Gr-I over a primary coat (Red oxide metal primer) (total three coats) of approved make and colour including cost of making the holes into RCC or masonry works and making good the same wherever necessary, including cost and conveyance of all materials, incidental charges etc., complete.	Supplying and fixing in position MS Ornamental grills for windows and vent	Supplying and fixing in position MS Ornamental gr	1002	141.03	1Kg	3490492.50

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17	6100.00	Supply & fixing of Hot dipped Galvanized Iron Chain link mesh 8 gauge (4mm) of size 2"x2" (50 mm x 50 mm) as per IS: 2721-2003 including the cost and conveyance and all incidental charges, etc., complete as directed during execution. The chain link mesh shall have a zinc coating of type 'heavy' with mass of coating 290 gm/sqm as per IS:4826-1979 (Reaffirmed 2001) Table.1. The mesh shall be provided with heavily coated galvanizede line wires of 4.50 dia. as given in Table.1 of IS:2721- 2003.And tying in position to the posts / angles or supports with GI wire including the cost of the G.I wire etc., complete as directed during execution and fixing with bolts and nuts to the angles.	Supply & fixing of Hot dipped Galvanized Iron Chain link mesh 8 gauge (4mm	Supply & fixing of Hot dipped Galvanized Iron Cha	NA	677.39	1Sqm	4132079.00
18	1915.00	Supply and fixing of Galvalume sheet Pre-painted Galvalume Trapezoidal Profile Roofing sheets with 0.47mm thickness (Total Coated Thickness-TCT), Coating: Alu-Zinc coating AZ150 GSM. Tensile strength: 550 MPA. Paint coating: Regular Modified Polyester painting. Painting thickness (Top): 18 to 20 Microns, (Bottom): 5 to 7 Microns. Sheet Width:1.02m, Length: Maximum 12 Meters with Regular Range Colours with minimum end lap of 15 cms etc., complete as directed during execution (Original coil shall be of make TATA Blue scope/ JSW)	Supply and fixing of Galvalume sheet Pre-painted Galvalume Trapezoidal Prof	Supply and fixing of Galvalume sheet Pre-painted G	NA	712.23	1Sqm	1363920.45
19	37.00	Supply of Roof Accessories medium size: Ridge, Barge Cap, Corner Trim, L-Flashing, Down Pipes, Gutter etc., with 0.47mm thick, Width: 0.585, Length: Maximum 2½ Meters	Supply of Roof Accessories medium size: Ridge, Barge Cap, Corner Trim, L-Fl	Supply of Roof Accessories medium size: Ridge, Bar	NA	318.56	1Rm	11786.72

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20	110.00	Painting two coats for new walls with Acrylic Emulsion paint exterior grade with silicon additives having VOC (Volatile Organic Compound) content less than 50 grams/ liter of approved brand and shade, over a priming coat with ready mixed cement primer (water thinnable) for exterior Grade-2 for new work (totally three coats), after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including brushes to have smooth finish, cost and conveyance of all materials to work site and all operational, incidental, labour charges etc., complete for finished item of work as per SS 911 for external walls.	Painting two coats for new walls with Acrylic Emulsion paint exterior gra	Painting two coats for new walls with Acrylic Em	911 1209 1201	296.27	1Sqm	32589.70
21	110.00	Painting two coats for new walls, with Acrylic Emulsion paint interior grade having VOC (Volatile Organic Compound) content less than 50 grams/ liter of approved brand and shade over a priming coat with Ready mixed cement primer (water thinnable) for interior grade-I for new work total three coats, after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc., complete for finished item of work as per SS 911 for internal walls.	Painting two coats for new walls, with Acrylic Emulsion paint interior gra	Painting two coats for new walls, with Acrylic Em	911 1209 1201	195.31	1Sqm	21484.10

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22	1950.00	Only labour charges for fixing precast concrete blocks (interlocking pavers) using not lessthan M40 grade paver blocks and thickness not less than 80 mm for medium traffic areas conforming to IS 15658-2006 of all shapes and designs as per the specification, over sand bed of 20 mm to 40 mm (30 mm average) thickness and compacting with vibrating plate compactor having 3 tonne compaction force there by forcing part of sand underneath to come up in between joints. Final compaction of paver surface to be done after spreading and brushing the fine sand into joints to its final level. The work includes cost of sand etc., labour and equipments such as cutting machine, lift, loading and unloading to work site clearing, bringing required all materials for the successful completion of work as per drawings and Technical Specification Clause 1504 MORD and as directed during execution by the Engineer-in-charge of the work. (Excluding cost of pavers)	Only labour charges for fixing precast concrete blocks (interlocking pavers	Only labour charges for fixing precast concrete bl	NA	330.45	1Sqm	644377.50

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23	1.00	Supplying, installation, Testing and commissioning of Electric engine driven main fire pump suitable for automatic operation and consisting of following: complete in all respect as required (a) End suction back pull out centrifugal pump, Std fitted with BRONZE IMPELLER & gland packing, capacity 137 Cum/hr (2280 LPM) at total head 70 mwc at operating speed 2900 rpm. Pump duty as per TAC (b) 45 KW / 60 HP, 3000 rpm, TEFC (IP55) horizontal foot mounted, squirrel cage induction motor, suitable for 415 volts +/- 10%, 3 Ph, 50 Hz +/-5%, a.c.supply. Class 'F' insulated and suitable for ambient temperature up to 50 deg C. Motor in frame SC 250 M frame. (c) M.S. Fabricated common base plate, coupling, coupling guard, foundation bolts etc as required. (d) Suitable cement concrete foundation duly plastered with anti-vibration pads.Makes of Pumps: Kirloskar / Mather&Platt/Lubi/KSB. Makes of Motors: Siemens/ ABB/ Kirloskar/Lubi/KSB	Supplying, installation, Testing and commissioning of Electric engine drive	Supplying, installation, Testing and commissioning	NA	279033.21	1 Set	279033.21

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24	1.00	Supplying, installation, Testing and commissioning of Diesel engine driven main fire pump suitable for automatic operation and consisting of following : complete in all respect as required (a) End suction back pull out pump, Std fitted with BRONZE Impeller & gland packing, capable of delivering 137 Cum/hr(2280 LPM) at a total head of 70 M when operated at 2150 rpm and pump duty as per TAC. (b) Liquid cooled, 68 BHP at 1800 rpm, Diesel engine (Radiator cooled) 12 Volts starting system with standard accessories. (c) M.S. Fabricated common base plate, coupling, coupling guard, foundation bolts etc as required. (d) Suitable cement concrete foundation duly plastered with anti-vibration pads. Makes of Engines : Kirloskar/Cummins. Makes of Pumps : Kirloskar / Mather&Platt/KSB.	Supplying, installation, Testing and commissioning of Diesel engine driven	Supplying, installation, Testing and commissioning	NA	566607.34	1 Set	566607.34

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25	1.00	Supplying, installation , testing and commissioning of electric driven pressurization / Jockey pump suitable for automatic operation and consisting of following : complete in all respect as required (a)Horizontal type, multistage, centrifugal pump of cast iron body & bronze impeller with stainless steel shaft, mechanical seal and flow of 180 lpm at 56 Mtr head conforming to IS 1520 (b)Suitable HP SQ cage induction motor TEFC type suitable for operation on 415Volts, 3 phase 50 Hz. AC with IP 55 class of protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-325 . (c) M.S. Fabricated common base plate, coupling, coupling guard, foundation bolts etc as required. (d) Suitable cement concrete foundation duly plastered with anti-vibration pads. Makes of Pumps : Kirloskar / Mather&Platt/KSB. Makes of Motors : Siemens/ ABB/ Kirloskar/KSB	Supplying, installation , testing and commissioning of electric driven pres	Supplying, installation , testing and commissionin	NA	69758.02	1 Set	69758.02

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26	1.00	Fabrication, supply, transportation, Installation testing & commissioning of Electrical control panel of cubical construction, floor mounted type, fabricated out of 2mm thick CRCA sheet, compartmentalized with hinged lockable doors, dust and vermin proof, powder coated of approved shade after 7 tank treatment process, cable alley, interconnection with suitable size copper conductor cable/solid copper strip, having switchgears and accessories, mountings and internal wiring, earth terminals, numbering etc. complete in all respect, suitable for main fire pump, pressurization pump & diesel pump set complete with following in coming and Outgoings, suitable for operation on 415V, 3 phase,50Hz Ac Supply. Makes: As per list of recommended brands. INCOMING : 250A, 50kA 4 Pole MCCB, Ics = 100%, Icu rating Digital Voltmeter 0-500V with selector switch Digital Ammeter (0-250 A) with selector switch & CTs etc. LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps Set of Copper Bus Bar 300A	Fabrication, supply, transportation, Installation testing & commissioning o	Fabrication, supply, transportation, Installation	NA	302438.55	Each	302438.55

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0	0.00	OUTGOING : (Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps) MAIN FIRE PUMP :125 A, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 60 HP pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation selector switch for local/remote, auto/manual/OFF operation. JOCKEY PUMP : 63 A, 50kA TPN MCCB, Ics=100% Icu, with suitable protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. DIESEL ENGINE CONTROL: Control for diesel engine comprising - Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode Indicating lamp for high/ low lube oil pressure, High Water Temp and Engine on indication Water Temp and Engine on indication. Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC voltmeter, and 0- 20 A DC Ammeter All standard relays and accessories for automatic operation of diesel engine System Controller.Makes Lauritz Knudsen (formerly L&T)- D shine / Schneider-NS NSX/Legrand -DP X3/ Siemens-3VL / Hager-h3/C&S - Winbreak2Makes of Meters : Schneider (Conzerve) / Lauritz Knudsen (formerly L&T) / Elmeasure / Siemens	OUTGOING : (Note : All outgoing feeders for pumps should have digital Amme	OUTGOING : (Note : All outgoing feeders for pumps	NA	0.00	NA	0.00

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27	0.00	Supplying, installing, testing and commissioning CI body Foot Valve with brass strainer with necessary flanges, nuts and bolts etc., complete on to the suction line. Makes:Normex /SANT/Kartar	Supplying, installing, testing and commissioning CI body Foot Valve	Supplying, installing, testing and commissioning C	NA	0.00	NA	0.00
(a)	3.00	- do - 150 mm nominal dia.	- do - 150 mm nominal dia.	- do - 150 mm nominal dia.	NA	17709.88	Each	53129.64
(b)	1.00	- do - 80 mm nominal dia.	- do - 80 mm nominal dia.	- do - 80 mm nominal dia.	NA	6788.30	Each	6788.30
28	0.00	Providing, installation, testing and commissioning of dual Plate non-return valve of following sizes confirming to IS: 5312 complete with rubber gasket, GI bolts, nuts, washers etc as required. Makes: BDK / NEU-G / Leader/ Zolotto / SANT/Honeywell//Kartar/Armorfire.	Providing, installation, testing and commissioning of dual Plate non-	Providing, installation, testing and commissioning	NA	0.00	NA	0.00
(a)	3.00	- do - 150 mm Dia.	- do - 150 mm Dia.	- do - 150 mm Dia.	NA	10714.13	Each	32142.39
(b)	1.00	- do - 80 mm Dia.	- do - 80 mm Dia.	- do - 80 mm Dia.	NA	5017.53	Each	5017.53
(c)	8.00	- do - 50 mm Dia.	- do - 50 mm Dia.	- do - 50 mm Dia.	NA	2921.42	Each	23371.36

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29	0.00	Manufacture, supply and delivery of following CI Wafer Lug type Butterfly Valves conforming to IS 13095/ 1991 (Reaffirmed 1998) excluding transportation, CED and taxes etc., complete. Operation done with worm actuator ISI marked. Makes: Audco / Kirloskar / BDK / H-Shankar / Leader/ Zolotto/SANT/NEU-G/Honeywell//Kartar/Armorfire.	Manufacture, supply and delivery of following CI Wafer Lug type Butterfly V	Manufacture, supply and delivery of following CI W	NA	0.00	NA	0.00
(a)	10.00	- do - 150 mm Dia.	- do - 150 mm Dia.	- do - 150 mm Dia.	NA	10330.95	Each	103309.50
(b)	1.00	- do - 80 mm Dia.	- do - 80 mm Dia.	- do - 80 mm Dia.	NA	5164.91	Each	5164.91
(c)	8.00	- do - 50 mm Dia.	- do - 50 mm Dia.	- do - 50 mm Dia.	NA	4132.15	Each	33057.20
30	4.00	Providing and fixing of Pressure Switch in the MS Pipeline including connection etc as required. Makes: Danfoss / Indfoss /System sensor/Honeywell.	Providing and fixing of Pressure Switch in the MS Pipeline including connec	Providing and fixing of Pressure Switch in the MS	NA	3227.50	Each	12910.00
31	4.00	Supply, installation, testing and commissioning of Bourdon type, stainless steel dial type pressure gauge with isolation valve and pipe having calibration of 0-16 kg/cm2. Makes: HGURU / FIEBIG	Supply, installation, testing and commissioning of Bourdon type, stainless	Supply, installation, testing and commissioning of	NA	860.44	Each	3441.76

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32	0.00	Supply and transportation of XLPE insulated, 1,100 V grade armoured aluminum cable as per specification confirming to IS:7098 (Part-I)/ 1554-I. Makes: Finolex / Paragon / V-Guard / Suncab / Delton / Payal / Laser /Rhino /Pawancab/AKG/LS cable/Polywires/Nakoda/South king	Supply and transportation of XLPE insulated, 1,100 V grade armoured aluminu	Supply and transportation of XLPE insulated, 1,100	NA	0.00	NA	0.00
(a)	20.00	- do - 16 Sq mm X 4 Core	- do - 16 Sq mm X 4 Core	- do - 16 Sq mm X 4 Core	NA	187.05	1Rm	3741.00
(b)	60.00	- do - 50 Sq mm X 3.5 Core	- do - 50 Sq mm X 3.5 Core	- do - 50 Sq mm X 3.5 Core	NA	412.65	1Rm	24759.00
(c)	100.00	- do - 120 Sq mm X 3.5 Core	- do - 120 Sq mm X 3.5 Core	- do - 120 Sq mm X 3.5 Core	NA	852.50	1Rm	85250.00
33	0.00	Supply, transportation & installation of following sizes of Hot Dipped Galvanized Iron perforated cable trays without cover, including horizontal and vertical bends, reducers tees, cross members and other accessories as required and duly suspended from the ceiling/Wall with suspenders etc., complete. (NB: Add 30% Extra for perforated cable tray with cover)	Supply, transportation & installation of following sizes of Hot Dipped Galv	Supply, transportation & installation of following	NA	0.00	NA	0.00
(a)	20.00	- do - 150mm x 50mm x 1.6mm	- do - 150mm x 50mm x 1.6mm	- do - 150mm x 50mm x 1.6mm	NA	564.56	1Rm	11291.20
(b)	30.00	- do - 300mm x 50mm x 1.6mm	- do - 300mm x 50mm x 1.6mm	- do - 300mm x 50mm x 1.6mm	NA	996.48	1Rm	29894.40

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34	30.00	Supply and laying of 25mm x 3mm Copper Flat (0.663Kgs/1Mtr) Earth wire/Strip in horizontal /Vertical run in Ground/surface/Recess including, riveting, soldering, saddles,making connection etc as required	Supply and laying of 25mm x 3mm Copper Flat (0.663Kgs/1Mtr) Earth wire/Stri	Supply and laying of 25mm x 3mm Copper Flat (0.663	NA	827.56	1Mtr	24826.80
35	0.00	Providing, laying, testing and commissioning of following C-Classes of MS Pipes conforming to IS 3589 and 1239 including fittings like elbows, tees,flanges, tapers, nuts, bolts, gaskets including transportation and painting with two (2) coats of red oxide paint and one coat of post office red enamel paint and all labour charges and cost of all accessories etc., complete. Makes: Jindal / Hissar / Tata pipes/Prakash Surya	Providing, laying, testing and commissioning of following C-Classes of MS P	Providing, laying, testing and commissioning of fo	NA	0.00	NA	0.00
(a)	270.00	- do - 150 mm dia	- do - 150 mm dia	- do - 150 mm dia	NA	4265.92	1Rm	1151798.40
(b)	50.00	- do - 100 mm dia	- do - 100 mm dia	- do - 100 mm dia	NA	2653.87	1Rm	132693.50
(c)	20.00	- do - 80 mm dia	- do - 80 mm dia	- do - 80 mm dia	NA	1761.69	1Rm	35233.80
(d)	24.00	- do - 65 mm dia	- do - 65 mm dia	- do - 65 mm dia	NA	1328.64	1Rm	31887.36
(e)	12.00	- do - 50 mm dia	- do - 50 mm dia	- do - 50 mm dia	NA	1062.23	1Rm	12746.76
(f)	22.00	- do - 40 mm dia	- do - 40 mm dia	- do - 40 mm dia	NA	900.12	1Rm	19802.64
(g)	116.00	- do - 32 mm dia	- do - 32 mm dia	- do - 32 mm dia	NA	710.80	1Rm	82452.80
(h)	212.00	- do - 25 mm dia	- do - 25 mm dia	- do - 25 mm dia	NA	531.68	1Rm	112716.16

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36	2.00	Supply, installation, testing and commissioning of 25 mm dia Ball valve. Makes : Zoloto / Leader / SANT/NEU- G/Kartar	Supply, installation, testing and commissioning of 25 mm dia Ball	Supply, installation, testing and commissioning of	NA	855.91	Each	1711.82
37	2.00	Supply, installation, testing and commissioning of 25 / 20 mm dia Air Release Valve. Makes : Zoloto / Leader / SANT/Kartar	Supply, installation, testing and commissioning of 25 / 20 mm dia Air	Supply, installation, testing and commissioning of	NA	1623.39	Each	3246.78
38	18.00	Supply and fixing single headed Internal/ External hydrant valve with instantaneous Stainless steel couplings of 63 mm dia with Cast iron wheel ISI marked conforming to IS 5290 (Type -A) with blank Gun Metal cap and chain as required - Single Headed Gun Metal. Makes: Newage/ Winco/ Padmini/Safex/Minimax/ Newtech/ Honeywell/ Kartar/ Armorfire/ Shuviro.	Supply and fixing single headed Internal/ External hydrant valve with insta	Supply and fixing single headed Internal/ External	NA	8633.88	Each	155409.84
39	18.00	Supplying and fixing of hose cabinet of size 750mm X 600mm x 250mm made of No 16 gauge SWG CRCA sheet with 6 mm thick glazed glass door including necessary locking arrangement suitable to accommodate 2 Nos 15 mtr long Hose pipe, 1 No branch pipe, mounted on wall OR raised brick platform and duly painted with Post office red externally and white internally with synthetic enamel paint complete in all respect, for external hydrant as required. Makes: Minimax/Safex/Newage/Winco /Padmini /NewTech /Kartar/Armorfire/Shuviro.	Supplying and fixing of hose cabinet of size 750mm X 600mm x 250mm made of	Supplying and fixing of hose cabinet of size 750mm	NA	4708.05	Each	84744.90

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40	18.00	Supplying and fixing 63 mm dia, 15 mtr long RRL hose pipe with 63 mm dia Male and Female Gun metal couplings duly binded with GI wire, rivets etc conforming to IS 636 (Type -A) as required. Makes: Newage/ Winco/ Padmini / Safex / Minimax /NewTech/Honeywell/Kartar/Armorfire/Shuviro.	Supplying and fixing 63 mm dia, 15 mtr long RRL hose pipe with 63 mm dia Ma	Supplying and fixing 63 mm dia, 15 mtr long RRL ho	NA	8309.65	Each	149573.70
41	18.00	Supply and fixing 63 mm dia Gun metal branch pipe with 20 mm (Nominal internal diameter) size Gun metal conforming to IS 903, suitable for instantaneous connection to interconnect hose pipe coupling as required. Makes: Newage/ Winco/ Padmini/Safex / Minimax/NewTech/Kartar/Armorfire/Shuviro.	Supply and fixing 63 mm dia Gun metal branch pipe with 20 mm (Nominal inter	Supply and fixing 63 mm dia Gun metal branch pipe	NA	3246.77	Each	58441.86
42	6.00	Supplying and fixing First -Aid Hose Reel with MS Construction spray painted in Post office Red, Conforming to IS 884 with upto date amendments, complete with the following as required. Makes: As per list of recommended brands. (a) 36.60 Mtr long 20 mm (Nominal internal) dia water hose Thermoplastic/ Textile reinforced/ Rubber braided/ Polypropylene Type - 2 as per IS:12585.Makes: Reliance/ Newage/ Safex /NewTech /Winco /Minimax/Kartar/Armorfire/Shuviro. (b) 20 mm (Nominal internal) dia gun metal ball/ gate valve and nozzle.Makes:NEU-G/SANT / Safex /Kartar/Armorfire/Shuviro. (c) Drum and brackets for fixing the equipments on wall. (d) Connections from riser with 25 mm dia stop valve (gun metal) & MS Pipe.	Supplying and fixing First -Aid Hose Reel with MS Construction spray painte	Supplying and fixing First -Aid Hose Reel with MS	NA	9887.70	Each	59326.20

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43	1.00	Supply and fixing 4 way fire brigade connection of cast iron body with 4 Nos Gun metal Male instantaneous inlet couplings complete with cap and chain as required for 150 mm dia MS Pipe connection, conforming to IS 904 as required. Makes: Newage/ Winco/ Padmini / Safex / Minimax/NewTech/Honeywell/Kartar/Armorfire/Shuviro	Supply and fixing 4 way fire brigade connection of cast iron body with 4 No	Supply and fixing 4 way fire brigade connection of	NA	11805.83	Each	11805.83
44	231.00	Providing fixing, testing and commissioning of 15mm size quartzoid bulb type sprinklers of rating 68º C. pendent / side wall with required accessories. Makes: HD / TYCO / VIKING / NEWAGE/ SHARP / SAFEX/Kartar/Armorfire/Shuviro.	Providing fixing, testing and commissioning of 15mm size quartzoid bulb typ	Providing fixing, testing and commissioning of 15m	NA	631.44	Each	145862.64
45	40.00	Supply and wrapping the anticorrosive material for buried pipes with coating of Primer and wrapping with 4 mm thick polymer corrosion resistant tape as per IS: 10221 with 15 mm overlap as per specifications - 100 mm nominal dia. Makes: IWL(PIPE KOT) or its equivalent.	Supply and wrapping the anticorrosive material for buried pipes with coatin	Supply and wrapping the anticorrosive material for	NA	399.04	1Mtr	15961.60
46	480.00	Supply and Transportation of XLPE/PVC insulated & PVC sheathed armoured cables of 1.1 KV grade with copper conductor as per IS 7098 Part-I -1988 etc., complete - 2CX1.5 Sq.mm. Makes: Torrent / Universal / Unicab / Havells / KEI /FINOLEX /Polycab/Gloster/Orbit/ Paragon/RR Kabel / KEC Asian Cables /Rhino /Pawancab/AKG/LS cable/ Polywires/South king/Avocab/Gemscab/Batra Henlay/PSPL / V-Marc.	Supply and Transportation of XLPE/PVC insulated & PVC sheathed armoured cab	Supply and Transportation of XLPE/PVC insulated &	NA	159.84	1Rm	76723.20

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47	1.00	Supply, install, test and commissioning of of 1 loop addressable peer to peer Networkable analogue addressable type fire alarm control panel with minimum 1500 Character / 240X64 Pixel / 7inch Touchscreen graphic LCD display. The panel should be equipped with sufficient numbers of loop with 20% spare capacity, with each networkable intelligent fire alarm Control Panel having SLC with capacity of min 250 detectors/ devices in any combination (Min 120 Detector) with key pad, dual flash-based microprocessor technology, inbuilt USB Port for upload and down load the configuration tools, RS232 serial port for direct PC or modem connection, inbuilt NAC's, min 20 programmable Zonal LEDs and operates on 240V AC + 10% with 50 Hz with built in charger.	Supply, install, test and commissioning of of 1 loop addressable peer to pe	Supply, install, test and commissioning of of 1 lo	NA	250961.77	Each	250961.77

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0	0.00	The panel shall have fire, fault relays, option of BMS integration MODBUS/ BACKNET, graphical software for provision of TCP/ IP modules, Remote Access, Class 'X' wiring with provision of both class 'A' and class 'B' wiring in the SLC circuit, user friendly with communication facility to monitor & control the FACPs from a single window. The panel shall be suitable for minimum of 5000 event record facility, event fetch from FACP facility and the single panel shall be suitable for min 2000 fire zones and 200 Panels/ Nodes in one network with all other accessories required to successfully run the system. The panel shall be completed as per UL and FM/EN54 & VDS standards (The System cost shall be included of all necessary cards, modules, Panel enclosure, CPU and associated accessories to complete the system design) Makes: Notifier / Edwards /Bosch /Tyco(simplex)/ Siemens- Desigo.	The panel shall have fire, fault relays, option of BMS integration MODBUS/	The panel shall have fire, fault relays, option of	NA	0.00	NA	0.00
48	10.00	Supply, Installation, Testing & Commissioning of Dust and Vermin proof (Resettable) addressable analogue Manual call boxes along with Isolators inbuilt having front reset mechanism with 170° viewable LED and can be viewed from the distance of 7 M. also have inbuilt isolator having operating temperature from 0°C to 49°C to initiate audio visual alarm in case of Human detection of fire. The manual call point must be indoor type. The call point must be resettable type without break glass. The item shall be completed as per UL (latest edition) and FM/EN54 & VDS standards. Makes: Morley / Honeywell/ Ravel /Seaara /Apollo /Cooper/Emkionyx	Supply, Installation, Testing & Commissioning of Dust and Vermin proof (Res	Supply, Installation, Testing & Commissioning of D	NA	4589.02	Each	45890.20

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49	10.00	Supply, installation, testing & commissioning of Analogue Addressable loop powered Sounder & Beacon with inbuilt isolator and having Min 15 distinct sound patterns/ multitone to indicate Exit doors and direct occupants for safe and fast evacuation and 100dB output with minimum 1Hz Frequency flash rate designed for IP65 requirements and approved to operate in 0°C to 49°C temperature, complies with the essential requirement of the EMC Directive to be supplied with Junction Box, Glands and other mounting accessories for proper installation.The item shall be completed as per UL (latest edition) and FM/EN54 & VDS standards. Makes: Siemens/Morley / Honeywell/ Ravel /Seaara /Apollo /Cooper/Emkionyx .	Supply, installation, testing & commissioning of Analogue Addressable loop	Supply, installation, testing & commissioning of A	NA	4245.52	Each	42455.20
50	11.00	Supply and fixing of ABC stored pressure squeeze grip type fire extinguishers, 9 kg capacity, IS:15683 : 2006, filled with ABC Mono Ammonium Phosphate base powder-90 complete with discharge hose and wall mounting bracket, 2mm Mild Steel, EPDM Rubbers, Pressure Guage, which work effectively on all class of fires (A, B, C & Electrical fires) with 5 years warranty Including transportation, all taxes and all labour charges etc complete.Makes: Safex / Kenex / Bharat / Reliance/NewTech/Kartar/Armorfire/Securezone/Shuviro.	Supply and fixing of ABC stored pressure squeeze grip type fire extinguishe	Supply and fixing of ABC stored pressure squeeze g	NA	6915.27	Each	76067.97

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51	11.00	Supply and fixing of Co2 type Fire Extinguishers 4.5 Kg capacity confirming to IS 2878 made from ISI marked steam less cylinder confirming to IS:7285 & CE certified and fitted with ISI marked controlled valve confirming to IS:3224, high pressure 1 Mtr discharge hose & horn complete with initial gas charged with carrying handle with wheels and wall mounting bracket. Squeeze lever discharge, used un-used indicator, epoxy coated paint (Red) with 93% gloss with 2 years warranty including transportation, all taxes and all labour charges etc complete. Makes: Safex / Kenex / Bharat / Reliance/NewTech/Kartar/Shuviro	Supply and fixing of Co2 type Fire Extinguishers 4.5 Kg capacity confirming	Supply and fixing of Co2 type Fire Extinguishers 4	NA	9354.88	Each	102903.68
52	2.00	Supply and fixing of 4 Nos of 9 Ltrs capacity round bottom sand buckets along with bucket sand, Fire buckets with round bottom type enamel painted, white inside & red outside and letter FIRE in black outside & Handle with mounting bracket.	Supply and fixing of 4 Nos of 9 Ltrs capacity round bottom sand buckets alo	Supply and fixing of 4 Nos of 9 Ltrs capacity roun	NA	1720.88	1 Set	3441.76
	0.00	**ELECTRICAL ITEMS**	ELECTRICAL ITEMS	ELECTRICAL ITEMS	NA	0.00	NA	**0.00**

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1	390.00	Supply and Fixing of ISI mark 25mm outer dia heavy grade (2.20mm thickness) with IS:9537 part -3 P.V.C. pipe for concealed wiring in walls of brick building duly cutting them to the required depth wherever necessary and fixing the pipes and specials with nails and plastering neatly with cement mortar and colour washing to match the finished surface and for surface wiring on chromium plated metallic base saddles on ceiling of roof / existing brick building walls, columns, beams, roofs / trusses / channel etc., and masonry work for light, fan and separate plug point including all labour charges etc., complete. Makes : Sudhakar / Finolex / Polycab / Kumar / AKG / Indiantubes / GM / Anchor / Nandi / Poly line / Goldmedal	Supply and Fixing of ISI mark 25mm outer dia heavy grade (2.20mm thickness)	Supply and Fixing of ISI mark 25mm outer dia heavy	NA	128.73	1Rm	50204.70
2	2.00	Supply and wiring with 2 run of 14/0.3mm (1.0 sq.mm) PVC insulated FRLS flexible copper cable (ISI MARK) as per IS:694 / 1990 specification in existing PVC pipes / casing & caping with pilot ceiling rose, 6 Amp 1 Way Module Modular switch to switch control box including all labour charges etc., complete for Light, bell, fan and Exhaust fan points in Residential Buildings. Grade-A Wire Makes : Finolex / RR kabel / Havells / KEI / Polycab / L&T Grade-B Modular switch Makes : Legrand Myrius /Crabtree Thames Platinum / Anchor Roma / Havells Coral or Fabio / Gold medal (Curve/Air) / Schneider Livia / C&S / L&T / HPL / GM (GNINE) / Kolors / Clipsal / Indoasian	Supply and wiring with 2 run of 14/0.3mm (1.0 sq.mm) PVC insulated FRLS fle	Supply and wiring with 2 run of 14/0.3mm (1.0 sq.m	NA	819.15	Each	1638.30

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3	85.00	Supply and run of 3 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible FRLS copper cable as per IS:694 / 1990 specification in the existing PVC pipe / MS pipe /PVC Trunking for mains including all labour charges etc., complete. Grade-A : Make: Finolex / RR kabel / Havells / KEI / Polycab / L&T	Supply and run of 3 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible F	Supply and run of 3 wire of 36/0.3mm (2.5 Sq mm)	NA	170.85	1Rm	14522.25
4	85.00	Supply and run of 4 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible FRLS copper cable as per IS:694 / 1990 specification in the existing PVC pipe / MS pipe /PVC Trunking for mains including all labour charges etc., complete. Grade-A : Make: Finolex / RR kabel / Havells / KEI / Polycab / L&T	Supply and run of 4 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible F	Supply and run of 4 wire of 36/0.3mm (2.5 Sq mm)	NA	215.81	1Rm	18343.85
5	85.00	Supply and run of 5 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible FRLS copper cable as per IS:694 / 1990 specification in the existing PVC pipe / MS pipe /PVC Trunking for mains including all labour charges etc., complete. Grade-A : Make: Finolex / RR kabel / Havells / KEI / Polycab / L&T	Supply and run of 5 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible F	Supply and run of 5 wire of 36/0.3mm (2.5 Sq mm)	NA	260.77	1Rm	22165.45
6	200.00	Supply and run of 6 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible FRLS copper cable as per IS:694 / 1990 specification in the existing PVC pipe / MS pipe /PVC Trunking for mains including all labour charges etc., complete. Grade-A : Make: Finolex / RR kabel / Havells / KEI / Polycab / L&T	Supply and run of 6 wire of 36/0.3mm (2.5 Sq mm) PVC insulated flexible F	Supply and run of 6 wire of 36/0.3mm (2.5 Sq mm)	NA	305.73	1Rm	61146.00

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7	36.00	Supply and fixing of 6A, 3-Plate Pilot Ceiling Rose on the existing junction boxes, T.W Board etc., and giving necessary connections. Makes: Anchor / Goldmedal (Olive) / GM (G Home) / Havells (Reo)	Supply and fixing of 6A, 3-Plate Pilot Ceiling Rose on the existing juncti	Supply and fixing of 6A, 3-Plate Pilot Ceiling Ros	NA	93.14	Each	3353.04
8	50.00	Supply and fixing of 16 /20 Amps 1 Way 1 Module Modular Switch on to the existing metal box (Switch Control Box) cover frame and giving necessary connections including all labour charges etc., complete. Grade B : Makes: Legrand Myrius /Crabtree Thames Platinum/ Anchor Roma / Havells Coral or Fabio / Gold medal (Curve/Air) / Schneider Livia / C&S / L&T / HPL / GM (GNINE) / Kolors / Clipsal / Indoasian	Supply and fixing of 16 /20 Amps 1 Way 1 Module Modular Switch on to the ex	Supply and fixing of 16 /20 Amps 1 Way 1 Module Mo	NA	123.53	Each	6176.50
9	6.00	Supply and fixing of 12 Modular Vertical Cover Plate & Frame on to the Existing Metal Box (Switch control Box) including all labour charges etc., complete. Grade- B : Makes: Legrand Myrius /Crabtree Thames Platinum/ Anchor Roma / Havells Coral or Fabio / Gold medal (Curve/Air) / Schneider Livia / C&S / L&T / HPL / GM (GNINE) / Kolors / Clipsal / Indoasian	Supply and fixing of 12 Modular Vertical Cover Plate & Frame on to the Exis	Supply and fixing of 12 Modular Vertical Cover Pla	NA	245.97	Each	1475.82
10	37.00	Fixing of M.V / S.V / M.H luminaire / Halogen fitting on wall / pole cross arm / channel / stem pipe etc., including all labour charges and giving connections etc., complete.	Fixing of M.V / S.V / M.H luminaire / Halogen fitting on wall / pole cross	Fixing of M.V / S.V / M.H luminaire / Halogen fitt	NA	285.18	Each	10551.66

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11	1.00	Supply and fixing of Three phase Electronic type energy Meter to operate in 415V, AC,50 Hz as per IS13010 Up to 40 Amps range with 200% over load capacity complete with connections etc., complete on the existing board. Make: CONZERVE / ELMEASURE / MECO / HPL / HAVELLS/L&T.	Supply and fixing of Three phase Electronic type energy Meter to operate in	Supply and fixing of Three phase Electronic type e	NA	3491.64	Each	3491.64
12	2.00	Supply and fixing of 125 Amp TP & N SDFU Switch as per IS:13947 with sheet steel enclosure without fuse units on wall etc., complete. Grade-A :Make: L&T / C&S / ABB / Siemens / GE/Schneider	Supply and fixing of 125 Amp TP & N SDFU Switch as per IS:13947 with sheet	Supply and fixing of 125 Amp TP & N SDFU Switch as	NA	13700.87	Each	27401.74
13	4.00	Supply and fixing of 4 Way TP & N Horizontal type Distribution board as per IS 8623; IS 13032; IEC 61439-3 with IP-43 protection (Metal door) firmly concealed in the wall with neat plastering etc., complete. Grade-A: Makes: Legrand (Ekinox 3) / Schneider-Acti9 / Hager - Novello+ / Seimens / Havells STADx / L&T-Newrang / Indo Asian-Opti Pro	Supply and fixing of 4 Way TP & N Horizontal type Distribution board as per	Supply and fixing of 4 Way TP & N Horizontal type	NA	4595.72	Each	18382.88
14	1.00	Supply and fixing of 4 Way VTPN D.Bs suitable for accomodating 125A / 160A 4 Pole MCCB as incomer and TP/SP MCBs as outgoing with IP-43 protection as perIS 8623; IS 13032; IEC 61439-3. (Metal door) firmly concealed in the wall with neat plastering etc., completed. [Excluding Incomer & Outgoings] Grade-A: Makes: Legrand(Ekinox 3) / Schneider-Acti9/Hager - Novello+/ Seimens/Havells STADx/ L&T-Newrang/Indo Asian(Opti Pro)	Supply and fixing of 4 Way VTPN D.Bs suitable for accomodating 125A / 160A	Supply and fixing of 4 Way VTPN D.Bs suitable for	NA	13681.92	Each	13681.92

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15	30.00	Supply and fixing of 6 to 32 Amps range SP MCBs 10 KA breaking capacity 'C'/ 'D' Curve ISI Mark as per IS:60898-1:2002 in the existing board and giving necessary connections. Grade-A: Makes: Legrand(Ekinox 3) / Schneider-Acti9/Hager - Novello+/ Seimens/Havells STADx/ L&T - Newrang / Indoasian (Opti Pro)	Supply and fixing of 6 to 32 Amps range SP MCBs 10 KA breaking capacity 'C	Supply and fixing of 6 to 32 Amps range SP MCBs 1	NA	358.38	Each	10751.40
16	4.00	Supply and fixing of 40 to 63 Amps range TP MCBs 10 KA breaking capacity 'C'/ 'D' Curve ISI Mark as per IS:60898-1:2002 in the existing board and giving necessary connections. Grade-A: Makes: Legrand(Ekinox 3) / Schneider-Acti9/Hager - Novello+/ Seimens/Havells STADx/ L&T - Newrang / Indoasian (Opti Pro)	Supply and fixing of 40 to 63 Amps range TP MCBs 10 KA breaking capacity '	Supply and fixing of 40 to 63 Amps range TP MCBs	NA	2137.03	Each	8548.12
17	4.00	Supply and fixing of 63A Four Pole, 30mA RCCB for earth leakage protection, confirms to IEC:61008 / IS: 12640 (Part-I) - 2008 in the existing boards including internal connections and Labour charges etc., complete. Grade-A: Makes: Legrand(Ekinox 3) / Schneider-Acti9/Hager - Novello+/ Seimens/Havells STADx/ L&T - Newrang / Indoasian (Opti Pro)	Supply and fixing of 63A Four Pole, 30mA RCCB for earth leakage protection	Supply and fixing of 63A Four Pole, 30mA RCCB for	NA	5090.19	Each	20360.76

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18	8.00	Fabrication, Transportation and supply of hot-dip galvanized Octogonal Pole with BSEN - 10025 grade S 355 JO steel plate for shaft, IS 2062 for base plate with HInged door opening arrangement, including the supply of suitable boards with bakelite sheet as per IS specification suitable to withstand the wind speed of 180 KMPH for 7.5 Mtrs height Pole having dimensions Bottom 130 mm,. Top 70mm with 3mm thick, base plate 220 x 220 x 12mm and 4 Nos of M20 x 700 long 'J' bolts along with template. Makes: Bajaj / Valmont/Laasma /Transrail/Utkarsh/ Skipper	Fabrication, Transportation and supply of hot-dip galvanized Octogonal Pole	Fabrication, Transportation and supply of hot-dip	NA	16790.49	Each	134323.92
19	8.00	Fabrication, Transportation and supply of 1.50 Mtr Length Single Arm Bracket made with MS galvanized 40mm / 50mm dia Pipe. (The design should be approved from field engineers) Make: Bajaj / Valmont/Laasma /Transrail/Utkarsh/ Skipper	Fabrication, Transportation and supply of 1.50 Mtr Length Single Arm Bracke	Fabrication, Transportation and supply of 1.50 Mtr	NA	2294.51	Each	18356.08
20	8.00	Foundation and erection charges with special T&P, for 7.50 Mtrs long Octagonal /Conical pole duly erecting on base plate duly providing 4 Nos foundation bolts with nuts with providing of 0.45Mtr x 0.45Mtrs x1.20 Mtr size M20 cc work vibrated concreting mixing with necessary steel (10kgs) reinforcement and curing for 7 days as directed by the field engineers during execution including excavation of Earth/BT all labour charges and cost and conveyance of materials etc.complete.	Foundation and erection charges with special T&P, for 7.50 Mtrs long Octago	Foundation and erection charges with special T&P,	NA	7766.64	Each	62133.12

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21	8.00	Supply & fixing of 1 1/4" dia 'B' class GI pipe not exceeding 1.5Mtr long with necessary bending (120° to 135°) to the wall or to the pole as directed during execution duly grouting 2 Nos. of MS angles of 35x35x5mm having length 20Cm (grouted portion of MS angle into the wall 7.5cm) and fixing with 2 Nos. of MS clamps fabricated with 25mmx3mm MS flat with suitable holes and 4 Nos. of 3/8"x2" MS bolts, nuts & washers for fixing on the wall / 2 Nos. of suitable MS clamps with bolts, nuts for fixing to the pole etc., complete. A 16Amp fuse unit is to be fixed to the MS angle / clamp and giving necessary connections with 14/0.3mm PVC copper flexible wire from pole to the fuse unit and to the fitting or from Ceiling rose in the building to the fuse unit and to the fitting with link clips etc., and the GI pipe has to be painted with two coats of enamel grey paint over one coat of red oxide paint.	Supply & fixing of 1 1/4" dia 'B' class GI pipe not exceeding 1.5Mtr long	Supply & fixing of 1 1/4" dia 'B' class GI pipe n	NA	1648.53	Each	13188.24

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22	1.00	Design, Fabrication, Supply, Installation and Commissioning of Outdoor type timer box of size 600mm(W) x 600mm(H) x 300mm(D) and made with 14SWG CRCA sheet steel cleaning with 7 tank process and painting with powder coating. The box shall be made by provide rubber beeding, door knobs etc., and the front door with double leaf arrangement with hinges and locking arrangement. Suitable heat Dissipation / Air / Ventilation ducts with inside wire mesh shall be provide to the panel. The box base shall be mounted on 40x40x6mm MS angle with stand legs at 4 corners at a length of 700mm each of 40x40x6mm MS angle with necessary welding and the angles shall be painted with two coats of enamel black paint over one coat of red-oxide. The timer box legs should be fixed in ground of size 400mm depth by using 20mm HBG metal in 1:3:6 ratio.	Design, Fabrication, Supply, Installation and Commissioning of Outdoor type	Design, Fabrication, Supply, Installation and Comm	NA	61921.31	Each	61921.31
23	1.00	Supply and fixing of suitable out door type feeder piller box made with 14 SWG CRCA sheet after 7 tank process including S & F of Legrand / Schnieder / L&T make 40-63A FP MCB, 9 Nos SP MCB's,din chanel, bus bars, 6 Sqmm 3 core copper cable for internal wiring , contactor, connectors and provision for energy meter etc., complete for finished item of work	Supply and fixing of suitable out door type feeder piller box made with 14	Supply and fixing of suitable out door type feeder	NA	25218.04	Each	25218.04
24	1.00	Supply and fixing of 205 X 255 X 112mm Sizes thermo plastic cable junction boxes with IP66/67 weather proof, for outdoor installation with konckouts for cable entry including 1 No. 6 to 32A SP MCB & terminals etc., complete (sizes in mm) Make : Hensel / Sadhrish / Sintex	Supply and fixing of 205 X 255 X 112mm Sizes thermo plastic cable junction	Supply and fixing of 205 X 255 X 112mm Sizes ther	NA	3288.68	Each	3288.68

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25	1.00	Supply and fixing of 255 X 600 X 170mm Sizes thermo plastic cable junction boxes with IP66/67 weather proof, for outdoor installation with konckouts for cable entry including 1 No. 6 to 32A SP MCB & terminals etc., complete (sizes in mm) Make : Hensel / Sadhrish / Sintex	Supply and fixing of 255 X 600 X 170mm Sizes thermo plastic cable junction	Supply and fixing of 255 X 600 X 170mm Sizes ther	NA	11110.87	Each	11110.87
26	170.00	Supply and transportation of 16 SqmmX 4 core XLPE insulated, 1100V grade armoured Aluminium UG cable as per specification confirming to IS:7098 (Part-I) / 1554-1. Grade-A: Make: Torent / Universal / Unistar / Havells / KEI / Gloster / Polycab / RR Kabel / Finolex/ Fortune Arrt	Supply and transportation of 16 SqmmX 4 core XLPE insulated, 1100V grade ar	Supply and transportation of 16 SqmmX 4 core XLPE	NA	251.67	1Rm	42783.90
27	30.00	Supply and transportation of 50 SqmmX 3.5 core XLPE insulated, 1100V grade armoured Aluminium UG cable as per specification confirming to IS:7098 (Part-I) / 1554-1. Grade-A: Make: Torent / Universal / Unistar / Havells / KEI / Gloster / Polycab / RR Kabel / Finolex/ Fortune Arrt	Supply and transportation of 50 SqmmX 3.5 core XLPE insulated, 1100V	Supply and transportation of 50 SqmmX 3.5 core XLP	NA	505.61	1Rm	15168.30
28	80.00	Supply and transportation of 120 SqmmX 3.5 core XLPE insulated, 1100V grade armoured Aluminium UG cable as per specification confirming to IS:7098 (Part-I) / 1554-1. Grade-A: Make: Torent / Universal / Unistar / Havells / KEI / Gloster / Polycab / RR Kabel / Finolex/ Fortune Arrt	Supply and transportation of 120 SqmmX 3.5 core XLPE insulated, 1100V grade	Supply and transportation of 120 SqmmX 3.5 core XL	NA	1055.43	1Rm	84434.40
29	200.00	Laying of U.G cables up to 50 Sq.mm through existing G.I.Pipe and back filling of trench including labour charges etc., complete.(Excluding the cost of G.I Pipe)	Laying of U.G cables up to 50 Sq.mm through existing G.I.Pipe and back fill	Laying of U.G cables up to 50 Sq.mm through existi	NA	51.73	1Rm	10346.00

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30	80.00	Laying of U.G cables from 70 Sq,mm to 150 Sq.mm through existing G.I.Pipe and back filling of trench including labour charges etc.,complete.(Excluding the cost of G.I Pipe)	Laying of U.G cables from 70 Sq,mm to 150 Sq.mm through existing G.I.Pipe a	Laying of U.G cables from 70 Sq,mm to 150 Sq.mm th	NA	102.27	1Rm	8181.60
31	180.00	Supply Laying and Transportation of Outer dia 50mm and inner dia 38mm, 6Kg/Sq.Cm (PE- 100 grade) double walled corrugated HDPE duct made as per specification BSEN - 500 86/IS 14930 Part-II. Make : DURA LINE (Dura Guard)/Precision/CPE or its equivalent make.	Supply Laying and Transportation of Outer dia 50mm and inner dia 38mm, 6Kg/	Supply Laying and Transportation of Outer dia 50mm	NA	161.47	1Rm	29064.60
32	50.00	Supply Laying and Transportation of Outer dia 78mm and inner dia 63mm, 6Kg/Sq.Cm (PE- 100 grade) double walled corrugated HDPE duct made as per specification BSEN - 500 86/IS 14930 Part-II. Make : DURA LINE (Dura Guard)/Precision/CPE or its equivalent make.	Supply Laying and Transportation of Outer dia 78mm and inner dia 63mm, 6Kg/	Supply Laying and Transportation of Outer dia 78mm	NA	200.01	1Rm	10000.50
33	50.00	Supply Laying and Transportation of Outer dia 160mm and inner dia 136mm, 6Kg/Sq.Cm (PE- 100 grade) double walled corrugated HDPE duct made as per specification BSEN - 500 86/IS 14930 Part-II. Make : DURA LINE (Dura Guard)/Precision/CPE or its equivalent make.	Supply Laying and Transportation of Outer dia 160mm and inner dia 136mm, 6K	Supply Laying and Transportation of Outer dia 160m	NA	381.39	1Rm	19069.50

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34	10.00	Manufacture, Supply and delivery of 150 mm dia NP-2 Class RCC Socket and spigot pipes conforming to B.I.S 458/2003 at Ex-factory (Rate per meter of effective length) excluding, Transportation Taxes & Duties. Make: As per B.I.S 458/2003	Manufacture, Supply and delivery of 150 mm dia NP-2 Class RCC Socket and sp	Manufacture, Supply and delivery of 150 mm dia NP-	NA	272.08	1Rm	2720.80
35	50.00	Supply and fixing of LT cable route indicators made out of cast iron fixed to one meter long 1"x1"x1/4" MS angle 4" bent at one end buried in the ground and the other end fixed with G.I. bolts and nuts to the cable route indicators as directed during execution	Supply and fixing of LT cable route indicators made out of cast iron fixed	Supply and fixing of LT cable route indicators mad	NA	666.23	Each	33311.50
36	80.00	Supply and making end termination with heavy Alluminium lugs (pin / ring type) upto 16 Sq.mm cable as per IS specification duly crimped with crimping tool, PVC tape etc., complete. Make: Dowels / Jainson / Comet.	Supply and making end termination with heavy Alluminium lugs (pin / ring ty	Supply and making end termination with heavy Allum	NA	15.87	Each	1269.60
37	16.00	Supply and making end termination with heavy Alluminium lugs (pin / ring type) for 50 Sq.mm cable as per IS specification duly crimped with crimping tool, PVC tape etc., complete. Make: Dowels / Jainson / Comet.	Supply and making end termination with heavy Alluminium lugs (pin / ring ty	Supply and making end termination with heavy Allum	NA	41.95	Each	671.20

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38	16.00	Supply and making one end termination with heavy duty single compression brass gland as per BS 6121:2005, IP 66 complete, SIBG type, heavy duty Alluminium lugs duly crimped with crimping tool, PVC tape etc for 50 Sq.mm with brass washers of Armoured PVC insulated & PVC sheathed / XLPE aluminium conductor cable of 1100 volt grade as required of size. Make: Dowels / Comet / SMI	Supply and making one end termination with heavy duty single compression br	Supply and making one end termination with heavy d	NA	264.14	Each	4226.24
39	16.00	Supply and making one end termination with heavy duty single compression brass gland as per BS 6121:2005, IP 66 complete, SIBG type, heavy duty Alluminium lugs duly crimped with crimping tool, PVC tape etc for 120 Sq.mm with brass washers of Armoured PVC insulated & PVC sheathed / XLPE aluminium conductor cable of 1100 volt grade as required of size. Make: Dowels / Comet / SMI	Supply and making one end termination with heavy duty single compression br	Supply and making one end termination with heavy d	NA	412.65	Each	6602.40
40	6.00	Supply and fixing of 12 Module Hot Dip Galvanized 20 / 18 SWG Module box with earthing terminal concealed in wall with all required accessories including masonry work and all labour charges etc., complete. Make: Legrand / G.M / L&T / GM Four-Five / Legrand Arteor / Schneider / Anchor / Clipsal / Havells / ABB / Indo Asian / Siemen / Fortune Art.	Supply and fixing of 12 Module Hot Dip Galvanized 20 / 18 SWG Module box wi	Supply and fixing of 12 Module Hot Dip Galvanized	NA	443.48	Each	2660.88
41	3.00	Providing of Independent Chemical Earthing after excavating an earth pit of size1.5 m X 0.9 m X 2.05 m(Top 1.50 m X 0.90 m X 0.9m & bottom 1.20 m X 0.90 m X 1.15 m) with the following specification:-	Providing of Independent Chemical Earthing after excavating an earth pit of	Providing of Independent Chemical Earthing after e	NA	8674.49	Each	26023.47

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	0.00	(a) The Earth Electrodes, 40mm dia, wall thickness 3.25 mm, terminal Size 25x8 mm, terminal hole dia 12 mm, 2 meter long, galvanized, including 1 bag (25Kg to a bag) of Earth Life Compound, as per IS 3043, Base Metal: IS marked branded mild steel pipes.	(a) The Earth Electrodes, 40mm dia, wall thickness 3.25 mm, terminal Size	(a) The Earth Electrodes, 40mm dia, wall thicknes	NA	0.00	NA	0.00
	0.00	Makes: Potential / Shake Hands / Mars / Tirumala / Arshlok earthing.	Makes: Potential / Shake Hands / Mars / Tirumala / Arshlok earthing.	Makes: Potential / Shake Hands / Mars / Tirumala /	NA	0.00	NA	0.00
	0.00	b) The Earth Electrode shall be inserted into the ground for 1.75 Mtrs. Below the inspection chamber bottom 300mm all around this Earth Electrode shall be covered with 25 Kgs. Of Earth Life Compound	b) The Earth Electrode shall be inserted into the ground for 1.75 Mtrs. Bel	b) The Earth Electrode shall be inserted into the	NA	0.00	NA	0.00
	0.00	(c) The excavated gap shall be back filled with excavated earth, and shall be properly consolidated with water.	(c) The excavated gap shall be back filled with excavated earth, and shall	(c) The excavated gap shall be back filled with ex	NA	0.00	NA	0.00
	0.00	(d) The top 0.25 m of Earth pipe shall be exposed into inspection chamber of Hume pipe . The all around inspection chamber shall be provided with 18" dia hume pipe. Earth connections shall be given to the Earth Electrodes with bolts, nuts & washers.	(d) The top 0.25 m of Earth pipe shall be exposed into inspection chamber o	(d) The top 0.25 m of Earth pipe shall be exposed	NA	0.00	NA	0.00

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	0.00	(e) A 65 mm dia PVC conduit is to be embedded through hume pipe at one side, to lay the Earth from the Earth bus to the Electrode through the PVC conduit and after laying the Earth, the pit entry PVC conduit is to be sealed with Bitumen compound.	(e) A 65 mm dia PVC conduit is to be embedded through hume pipe at one sid	(e) A 65 mm dia PVC conduit is to be embedded thro	NA	0.00	NA	0.00
	0.00	(f) Necessary Meggar testing is to be done in the presence of the concerned Engineers, and the reading should be as per IE standards.	(f) Necessary Meggar testing is to be done in the presence of the concerned	(f) Necessary Meggar testing is to be done in the	NA	0.00	NA	0.00
42	30.00	Supply and run of 25 mm X 3 mm G.I Strip for earth electrode to required places by concealing in walls, flooring by cutting the grooves and replastering neatly and giving necessary connection with 3/8"x2" GI Bolts, Nuts and Washers wherever joints has to be done. The strip has to be either riveted (or) bolted (or) gas welding neatly and all labour charges etc., complete.	Supply and run of 25 mm X 3 mm G.I Strip for earth electrode to required	Supply and run of 25 mm X 3 mm G.I Strip for ea	NA	158.55	1Rm	4756.50
43	108.00	Earth Work excavation for trench for laying of U.G. Cable in ordinary rock (not requiring blasting) to the required width and depth and depositing the same on bank as directed during execution etc., complete	Earth Work excavation for trench for laying of U.G. Cable in ordinary rock	Earth Work excavation for trench for laying of U.G	NA	501.07	1Cum	54115.56

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44	2.00	Construction of UG cable Man hole of clear size 1 Mtr X 0.6 Mtrs X 0.9 Mtrs is to be constructed with sand filling 10mm and laying Cement Concrete (1:4:8) with 10mm twice using 40mm HBG metal and construction of 9" brick masonry wall with C.M 1:6 and plastering in cement motor 1:5 inside / out side and bottom of man-hole and including cost and laying charges of precast slab of 50mm thick over manhole including cost of steel and fabrication charges including cost and conveyance of all materials, curing and incidental charges etc., complete Note: Excluding the cost of Earth Work	Construction of UG cable Man hole of clear size 1 Mtr X 0.6 Mtrs X 0.9 Mtrs	Construction of UG cable Man hole of clear size 1	NA	10235.56	Each	20471.12

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45	2.00	Supply and Fixing of 4 feet T5, 18W LED Batten type fitting with Internal driver with Polycarbonate housing & PC diffuser suitable to operate on 230V AC, 50 Hz power supply as per the following specification. Make: Philips / Osram / Havells / Crompton Greaves / GE / Orkus. Specifications: (1). Input Power: 18W ±5%. (2). Input Voltage and Frequency: 230V ±5% AC, 50Hz. (3). Power factor : >0.95. (4). THD % : <15% (5). Operating Voltage : 140 - 270 VAC (6). Surge Protection : ≥ 2.5 kV (7) LED make: Nichia (Japan) / Osram (German) / Philips Lumiled / Cree (USA) / Bridgelux (USA) / Seoul / Lednim / Everlight / Epistar / Edison. (8). Luminous Flux (Lumen) : 1800 ± 5% (9) Luminous Efficiency (LM/W):100±5% (10) Colour Temperature (K): 5500K to 6500K (11) CRI : ≥ 80; (12) LED Beam Angle : >150º (13) IP Grade: IP20; Warranty : 5 Years.	Supply and Fixing of 4 feet T5, 18W LED Batten type fitting with Internal d	Supply and Fixing of 4 feet T5, 18W LED Batten typ	NA	241.00	Each	482.00

S.No.	Estimate Quantity (only Figures)	Item Detailed Specification Description	Work Type eg. Earth Work, Electrical works.. etc (upto 200 Characters)	Item Short Description (upto 100 Characters)	APSS / Morth Cl. Number (upto 200 Characters)	Rate (INR) (Upto 2 Decimals)	UOM (upto 50 Characters)	Amount (INR) (Upto 2 Decimals)
46	27.00	Supply and transportation of 60W LED High Bay Luminaire with high power COB LED made of pressure diecast aluminum housing with powder coated, supply input voltage 120 - 270 V AC, P.F > 0.90, high power LEDs, having system efficacy >=120 lumens/watt and junction temperature < 70°C, with ingress protection IP66, IK05, Driver surge protection 4KV, in-house additional surge protection ≥ 5KV with wide/narrow beam optics, THD<10% at 110 Volts AC, driver efficiency >90%, , CCT: 3000K - 5700K, minimum CRI>70, etc., complete with earth connection and not less than two years warranty a) LUMINAIRE MAKE : Wipro / Philips / GE-Venture / Crompton/ Bajaj /Greenlites / Havells/ /JLCJustled/Capart(Premium) / Sturlite/ Halonix/ Polycab / Keselec/ HPL / Jaquar / Eveready / Surya (ORION-II)/ FortuneArrt (UFO-S) / Rhino(GPHB02 Series) / Ecolink(Megalux) / Bee-Lite / Transrail. b) LED MAKE : PHILIPS LUMILEDS / CREE / NICHIA / OSRAM / SAMSUNG	Supply and transportation of 60W LED High Bay Luminaire with high power COB	Supply and transportation of 60W LED High Bay Lumi	NA	5676.19	Each	153257.13

S.No.	Estimate Quantity (only Figures)	Item Detailed Specification Description	Work Type eg. Earth Work, Electrical works.. etc (upto 200 Characters)	Item Short Description (upto 100 Characters)	APSS / Morth Cl. Number (upto 200 Characters)	Rate (INR) (Upto 2 Decimals)	UOM (upto 50 Characters)	Amount (INR) (Upto 2 Decimals)
47	10.00	"Supply and transportation of 60W LED Street light Luminaire made of pressure diecast aluminum body with powder coated, having protective toughened glass/ temperature protective polycarbonate diffuser, supply input voltage 120 - 270V AC, P.F > 0.90, high power LEDs having System Efficacy >=120 lumens/watt and junction temperature < 70°C, with Ingress protection IP66, Luminaire performance complies to IS 10322 (Part 5/Sec-3), driver surge protection 4KV, in-house additional Surge protection 10 KV with optics distribution, THD<10% at 110 Volts AC, driver efficiency >90%, CCT: 3000K - 5700K, minimum CRI>70, etc., complete with earth connection and not less than two years warranty. a) LUMINAIRE MAKE :Wipro(Skyline) / Philips (Green line) / GE-Venture / Crompton(Neo Series) / Bajaj (Edge) / Halonix(Lumos - Super) / Havells (Endura Series) / Valmont(Supernova) /JLCJustled (Bat series) / Capart (Premium) /Greenlites(Hi-Lux) /Sturlite(Asteor) /Polycab / Keselec/ HPL(City Vision)/ Jaquar(Premium) /Eveready/ Surya (Lexon) / FortuneArrt (Leaf) / Rhino(GPSL04) / Orient(Fiest Plus)/ Bee-lite(Gritty) / Ecolink(Durastreet Gen II) /Pro-Lite(GLM)/Transrail(Ursa minor)."	"Supply and transportation of 60W LED Street light Luminaire made of pressu	"Supply and transportation of 60W LED Street light	NA	3940.57	Each	39405.70

S.No.	Estimate Quantity (only Figures)	Item Detailed Specification Description	Work Type eg. Earth Work, Electrical works.. etc (upto 200 Characters)	Item Short Description (upto 100 Characters)	APSS / Morth Cl. Number (upto 200 Characters)	Rate (INR) (Upto 2 Decimals)	UOM (upto 50 Characters)	Amount (INR) (Upto 2 Decimals)
48	1.00	Design, fabrication, supply and errection of L.T indoor type cubical panel board made with 14 SWG CRCA sheet. The size of panel should be 1350(w)mm x 1500(H)mm x 400(D)mm should be mounted on base ISMS MS channel size 100x50x6mm and fixing firmly as directed execution. The following switch gears, bus bars etc., should be supply and fixing in the panel firmly, and the panel with necessary individual doors, separate chambers for bus bars, out going cables etc. All the doors shall be provided with rubber beading and door knobs. The Top of the box should have provision for Incoming and Outgoing cables. The entire panel should be powder coated with siemens grey color through 7 tank process.	Design, fabrication, supply and errection of L.T indoor type cubical panel	Design, fabrication, supply and errection of L.T i	NA	314701.24	Each	314701.24
	0.00	Incoming: (1). 1No. of 400A 4 Pole MCCB, adjustable, confirms to IEC 60947-2 and IS 13947-Part 1 & 2, having Breaking Capacity 36/35KA with thermal magnetic release Panel Mounted. Makes: Lauritz Knudsen (Formerly L&T)- D shine / Schneider-CVS / Legrand –(DP X3 / DRX) / Siemens- 3VL / Hager-h3 / C&S -Winbreak / HPL-Techno Outgoing: (1) 5Nos. of 250A 4 Pole MCCB, adjustable, confirms to IEC 60947-2 and IS 13947-Part 1 & 2, having Breaking Capacity 36/35KA with thermal magnetic release Panel Mounted. Makes: Lauritz Knudsen (Formerly L&T)- D shine / Schneider-CVS / Legrand –(DP X3 / DRX) / Siemens- 3VL / / Hager-h3 / C&S -Winbreak / HPL-Techno	Incoming: (1). 1No. of 400A 4 Pole MCCB, adjustable, confirms to IEC 60947-	Incoming: (1). 1No. of 400A 4 Pole MCCB, adjustabl	NA	0.00	NA	0.00

S.No.	Estimate Quantity (only Figures)	Item Detailed Specification Description	Work Type eg. Earth Work, Electrical works.. etc (upto 200 Characters)	Item Short Description (upto 100 Characters)	APSS / Morth Cl. Number (upto 200 Characters)	Rate (INR) (Upto 2 Decimals)	UOM (upto 50 Characters)	Amount (INR) (Upto 2 Decimals)
	0.00	Copper Bus bar arrangement: (1) 4 Nos of size 2" x 1/4" copper flat shall be run from 400A MCCB unit to vertical bus bar chamber with 400x12x50mm insulators supports and shall be run vertically in Bus bar chamber fix the bus bar with necessary insulators supports. (2) 4 Nos of size 1" x 1/4" copper flat links shall be provided to each 250 Amps MCCB connect to main bus bar and provide extension bits at out going side for cable connection with necessary bolts and nuts and using necessary 40x8mm insulator supports. (3) 1 No. of size 1" x 1/4" copper flat shall be run inside bottom width of the panel for earth bus bar. (4) Necessary heat shrinkable sleeve should be provided to the above copper bus bar except earth flat. (5) 2 Nos. of danger boards to be fixed to the board.	Copper Bus bar arrangement: (1) 4 Nos of size 2" x 1/4" copper flat shall b	Copper Bus bar arrangement: (1) 4 Nos of size 2" x	NA	0.00	NA	0.00
	0.00	Metering: (1) Supply and Fixing of energy analyser (multy function energy meter) with LCD/LED display capable of minimum 20 numbers electrical parameters display with 1A/5A CT input for LV/ HV network, RS 485 communication port etc.c, omplete in all respect - 1No. (2) Supply and Fixing of 240/220 V, LED (22.5 mm dia) Pilot lamp with integral circuit, terminal block, including connection etc. as required - 1Set	Metering: (1) Supply and Fixing of energy analyser (multy function energy m	Metering: (1) Supply and Fixing of energy analyser	NA	0.00	NA	0.00
		Rupees Three crores thirty four lakhs fourteen thousand seven hundred thirty two and paise forty six only						**3,34,14,732.46**

Digitally signed by
MANOHARAM BADDELI
Date: 22-10-2026
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